User manual version: 1.0
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Preliminaries

The following user manual is intended to guide the other end-users of the *Biogoals.TE-LP* tool.
This tool is intended to be used by biomethane plant managers decision-makers. It has the aim to provide a preliminary techno-economic optimisation of the input diet of a digester, considering some a linear simplified formulation of the net profit of the latter and some linear constraints to the input diet mix itself. In particular, if the co-feedstocks' purchase and transportation costs from the suppliers in the nearby area are known or can be hypothesised, it suggests **WHAT to buy from WHO** (see "**RobustOptimization_Project_Proposal.pdf**" for further contextual details).

Given some input data about each feedstock's availability on the nearby territory, its purchase/transportation cost and main biochemical characteristics (e.g. biomethane potential i.e. BMP, volatile solids' content etc...), the program solves a linear optimisation problem ('deterministic' LP) and returns the **diet mixture that may lead to the highest operational profit possible, while guaranteeing that some technical and regulating-authority constraints are satisfied** (e.g. GSE rules to access incentives or digestate quality).
Since some input data are usually uncertain in the industrial practice, if more than one realisation of the same data or an hypothesis on the uncertainty range is available, the program solves a **robust linear optimisation problem** ('robust' LP with the *Chance-constraint* approach). Setting the ***risk level*** **the user is willing to undertake in face of the uncertain input data**, the optimal safer diet mixture is returned i.e. the diet that maximises profit in the case the worst realisation among the ones considered within the desired risk level comes true in real life (see "**RobustOptimization_Project_Report.pdf**" for mathematical details).

Files description

- (MAIN) ***"/Project/Biogoals_TELP.ipynb"***: main optimization code, program, Jupyter notebook.
- ***"/Project/my_utilities.py"***: custom utility functions used in MAIN, Python file.
- ***"/Project/bmp_model.py"***: functions with BMP kinetics model (two-pool first-order) to be used in MAIN, Python file.
- ***"/Project/Project_database.xlsx"***: Excel input data file (costs, availability, characteristics).
- ***"/Project/RobustOptimization_Project_Report.pdf"***: report produced for the PhD course of Politecnico di Milano with reference to the example case-study present in the code.

- ***"/Project/RobustOptimization_Project_Proposal.pdf"***: context and introduction produced for the PhD course of Politecnico di Milano with reference to the example case-study present in the code.
- ***"/Project/Results.xlsx"***: Excel output file (solutions of the deterministic and robust problems for different hydraulic retention times (HRT) and risk levels (`Prob`)).
- ***"MATH80624_LectureNotes.pdf"***: lecture notes of Professor Erick Delage with reference to the mathematical formulations of the robust problem and uncertainty sets design.

"Project_database.xlsx"

- **Which input data are needed to be collected?**
  - **Feedstocks' characteristics**: for each co-feedstock of interest, specify:
    - **'rho'**: density;
    - **'TS'**: total solids content;
    - **'VS'**: volative solids content;
    - **'BMPinf'**: ultimate biochemical methane potential;
    - **'Fixed_tariff'**: fixed part of the transportation cost (usually expressed in terms of € for every 20 tons of feedstocks' fresh matter);
    - **'TKN'**: Total Kjedahl Nitrogen content;
    - **'C/N'**: molar carbon over nitrogen content ratio;
    - **'TAN'**: Total Ammoniacal Nitrogen content;
    - **'TAC'**: Total Alcalinity content in bicardonate equivalent;
    - **'LI'**: Lipid content;
    - **'TVFA'**: sum of volatile fatty acids (acetic, propionic, butirric and valeric acids) content in acetic acid equivalent;
    - **'COD/VS'**: total chemical oxygen demand over VS ratio;
    - **'k_hyd_r'**: hydrolysis constant of the first days of the BMP experiment (readily biodegradable part) in days;
    - **'k_hyd_s'**: hydrolysis constant of the last days of the BMP experiment (slowly biodegradable part) in days;
    - **'BMPinf_r'**: biochemical methane potential value reached in the first days of the BMP experiment (readily biodegradable part);
    - **'BMPinf_s'**: biochemical methane potential value reached between the end of the first days of the BMP experiment and the end of the experiment itself (slowly biodegradable part);
    - **'Leg. Cluster'**: to be consistent with the legislation (Italian DM 23 giugno 2016) about the composition of the input diet mixture to access the incentive scheme, each feedstock has to be catalogued (i.e. `Legislative Cluster` and `itemtype`) as:

        - 'product' (P);
        - 'by-product' (B);
        - 'animal slurry' (S);
        - 'second harvest crop' (P2).

          It is possible to extend this also to 'wastes' (W). The limits on the diet mixture classification and relative limitations (later implemented in the optimisation problem, see the 'Biogoals_TELP.ipynb' section of this manual) were taken from this link.
    - Some characteristics can be uncertain, if you have only a min-max range, the above-mentioned value is the mean. For this example, the only uncertain data reported were TS, VS, BMP (you should report in a proper column both min and max values i.e. `TS Max`, `TS Min`, `VS Max`, `VS Min`, `BMPinf Max`, `BMPinf Min` columns).
    - Note: `k_hyd_r`, `k_hyd_s`, `BMPinf_r`, `BMPinf_s` are not needed if a real complete BMP curve is available for each feedstock, since the latter can be used directly in the code.
  - **Distance** of each possible **supplier** from the plant of interest.
  - **Availability** of each feedstock of each possible supplier over the purchase horizon of interest (one year for the example provided). The overall amount can be uncertain (but not the repartition of the overall amount to each supplier i.e. the *size*/production capacity of the suppliers is known): Mean, Max, Min are provided; Max/Mean and Min/Mean are computed.
  - **Purchase cost** of each feedstock uncertainty. The market values are considered, i.e. equal for all suppliers. Actually, in the deterministic problem, different prices for the different suppliers can be considered; however, in the current approach it is not possible to consider different uncertainty ranges for each supplier separately in the robust counterpart. The market values can be uncertain: Mean, Max, Min are provided; Max/Mean and Min/Mean are computed.

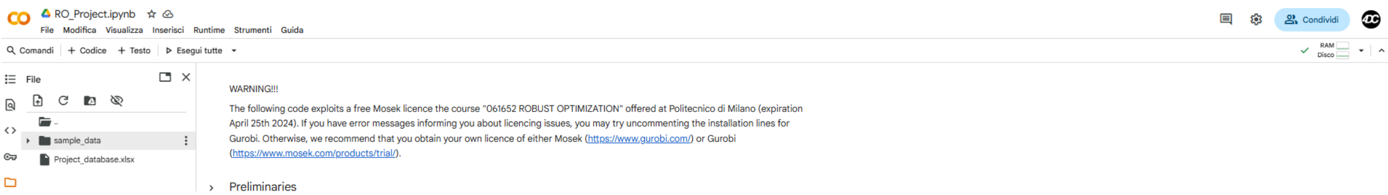
In the example provided, the `z` supplier is the farm where the biomethane plant of interest is located. The transportation costs for the feedstocks available at `z` are thus null. Yet, the purchase cost is not null as it is considered that the feedstocks produced in the farm can be sold anyway to third parties. It is possible or not to add a constraint in the problem formulation to force the consumption of the feedstocks produced *in loco* (see the 'Biogoals_TELP.ipynb' section of this manual).

"Biogoals_TELP.ipynb"

Solvers: the project was carried out with a *Mosek* academic licence offered by Politecnico di Milano. *Gurobi* is also suggested as a solver compatible with the *rsome* package. You can find the list of the compatible solvers here. The installation of solver packages is recommended with Anaconda (via '*conda*'), instead of via '*pip*'.
If you have problems with local Python and/or Jupyter runtime, a run-ready version of the same Jupyter notebook is available at this **COLAB** link.

After the installation of the ***rsome*** package, standard libraries are imported as well as some custom functions ('.py') present in the same directory. Afterwards:

- Load the input data: SPECIFY THE PATH TO THE INPUT DATA EXCEL FILE. If Google Colab is used, upload the file in the Colab local directory (see figure below).

  Re-declare or compute some of the values read (e.g. compute the matrix of transportation costs for each supplier and each feedstock).
- SPECIFY THE PARAMETERS OF THE PROBLEM e.g. number of co-feedstocks, plant size, constraint's (upper and lower) bounds over some quantities computed linearly from the diet (e.g. volatile solids (VS), total ammonical nitrogen (TAN), total volatile fatty acids (TVFA) of the input diet and HRT of the plant). See the user manual on feedstocks' characterization for other information on how to compute such quantities for each co-feedstock if not already available. Attention: the `N` constant specified in the python file must be coherent with these amounts…in this case, yearly availability!
- Create the whole BMP time curve from the parameters read (from the hydrolytic constants k_{hyd} and the cumulative biomethane potential BMP$_{\infty}$ the time series of batch production potential at each day of the test is extracted). Nota that if complete real BMP curve data are available to decision-makers, this step is not needed and it is recommended to directly load the experimental time-series data!
- **SOLVE THE DETERMINISTIC PROBLEM** for each user-fixed HRT value (within the pre-defined bounds HRT$_{min}$ and HRT$_{max}$)...SPECIFY THE HRT VALUE, run for all the HRT values of interest and results are saved to Excel. Compute some quantities and save to Excel...SPECIFY THE PATH TO THE EXCEL OUPUT FILE. If Google Colab is used, upload the file in the Colab local directory as previously shown.
- **INTRODUCTION OF UNCERTAINTY**0 on each feedstocks':
  - BMP (no TS and VS for simplicity);
  - purchase cost;
  - overall availability over the purchase horizon (fixed re-partition of the overall uncertain availability to the different suppliers i.e. hypothesis of fixed supplier 'size').
    Since many historical data were not available in the current example (but they could and should be in the industrial practice), some hypothesis were made regarding the *a priori* distribution of such uncertain quantities. Random realisations can be generated within the (min,max) values via the *samples* function (for purchase costs, the whole time series were randomly recreated considering that agri by-products/products market costs are updated

once per week. 5 years of data were considered). BMP distributions were instead hypothesised to be symmetric so that a theoretical result could be exploited (more important the 'protection' against unexpected small BMP values). At this point, the dimension of the uncertainty sets (i.e. how conservative I want to be with my diet against uncertainties) for each form of uncertainty are computed based on the desired risk level (`Prob` $\in [0,1]$) SPECIFY `Prob`

- **SOLVE THE ROBUSTIFIED PROBLEM** for each user-fixed HRT value. In the code example, it is possible with two boolean parameters (i.e. `uncertain_BMPs` and `uncertain_costs`) to DECIDE WHICH UNCERTAINTY SOURCE IS ACTIVE. To write down the robust counterpart of the deterministic problem solved above, the first thing is to CHOOSE THE GEOMETRY OF THE UNCERTAINTY SETS (e.g. box, ellipsoidal, budgeted) to write down the constraint on each uncertain variable. SPECIFY THE HRT VALUE, run for all the HRT values of interest and results are saved to Excel. See "RobustOptimization_Project_Report.pdf" for more details about the mathematical formulation of the objective function and constraints (when uncertainty is present in the objective function, it is better to re-write it as constraints, as it was done in the example). Compute some quantities and save to Excel...SPECIFY THE PATH TO THE EXCEL OUPUT FILE.
- Since in our case the value of the objective function is directly impacted by the uncertainty in feedstocks' purchase cost and BMP, the deterministic and robust solutions can be **TESTED OVER SOME POSSIBLE FUTURE ACTUAL REALIZATIONS of data**. Random realisations for the next 20 years of plant operations (i.e. 20 purchase horizons) are created and the objective function is evaluated considering these costs and the solution provided with the uncertainty sets previously designed. This test can be done for each HRT value defined by the user. It is important to check that the `R-LP J value` (maximum robust i.e. guaranteed profit) is below the resulting box plot of tested J values, but not too much (symptom of overly conservativeness). If the `R-LP J value` is too far from the testing box plot, it is recommended to decrease the dimension of the uncertainty set and try solving the robust problem again, iteratively, and viceversa. In the code, an example with random realisations of the feedstocks' purchase costs is present (note that, since in this example only costs uncertainty is considered for testing, also the `R-LP J value` used for comparison must be found solving the robust problem with with only `uncertain_costs = True` and `uncertain_BMPs = False`).
- After the user have run the above-mentioned procedure for the `Prob` and HRT values of its interest, the same decision-support plots present in "**RobustOptimization_Project_Report.pdf**" can be obtained. SPECIFY THE PATH TO THE EXCEL FILE, the same where the output have been previously stored.

Additional notes

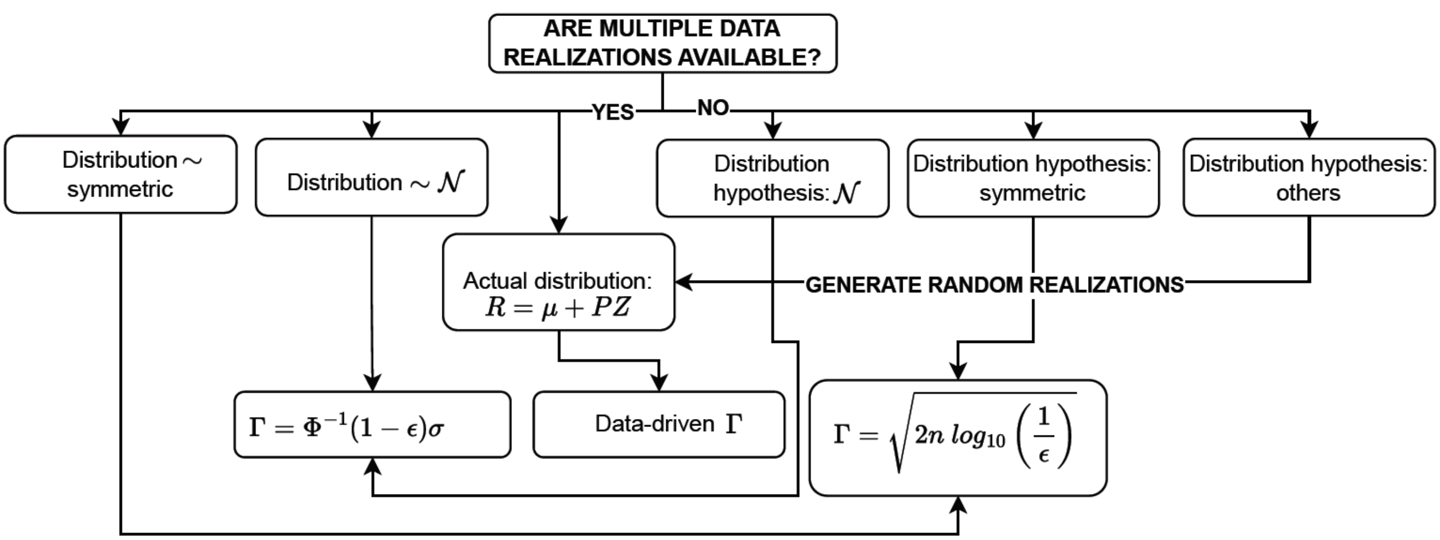
Regulating authority constraints

The limits imposed by the regulating authority on the diet mixture to access the incentives are:

- $\sum cultures\ of\ Table\ 1B < 30\%$
- $\sum 2nd - turn\ cultures < 20\%$
- $\sum slurries\ and\ stocks\ (byproducts\ of\ Table\ 1A) > 70\%$
- $\sum wastes = 0\%$ (not directly implemented in the problem)
- $\sum delicate\ cultures < 30\%$ (not implemented in the problem)
  *tables 1A and 1B specified in the Italian DM 23 giugno 2016.
  To mimic the presence of additional limitations imposed by the regulating authority about digestate quality and composition, ad additional constraint over the maximum residual BMP$_{res,max}$ was introduced.
  $\sum BMP_{res} < 15\%\ BMP(HRT)\ of\ the\ input\ diet$
  Similarly, on many others digestate quantities can be introduced if the value of such quantify is added in the Excel input data and the diet-weighted average is computed in the constraint formulation.

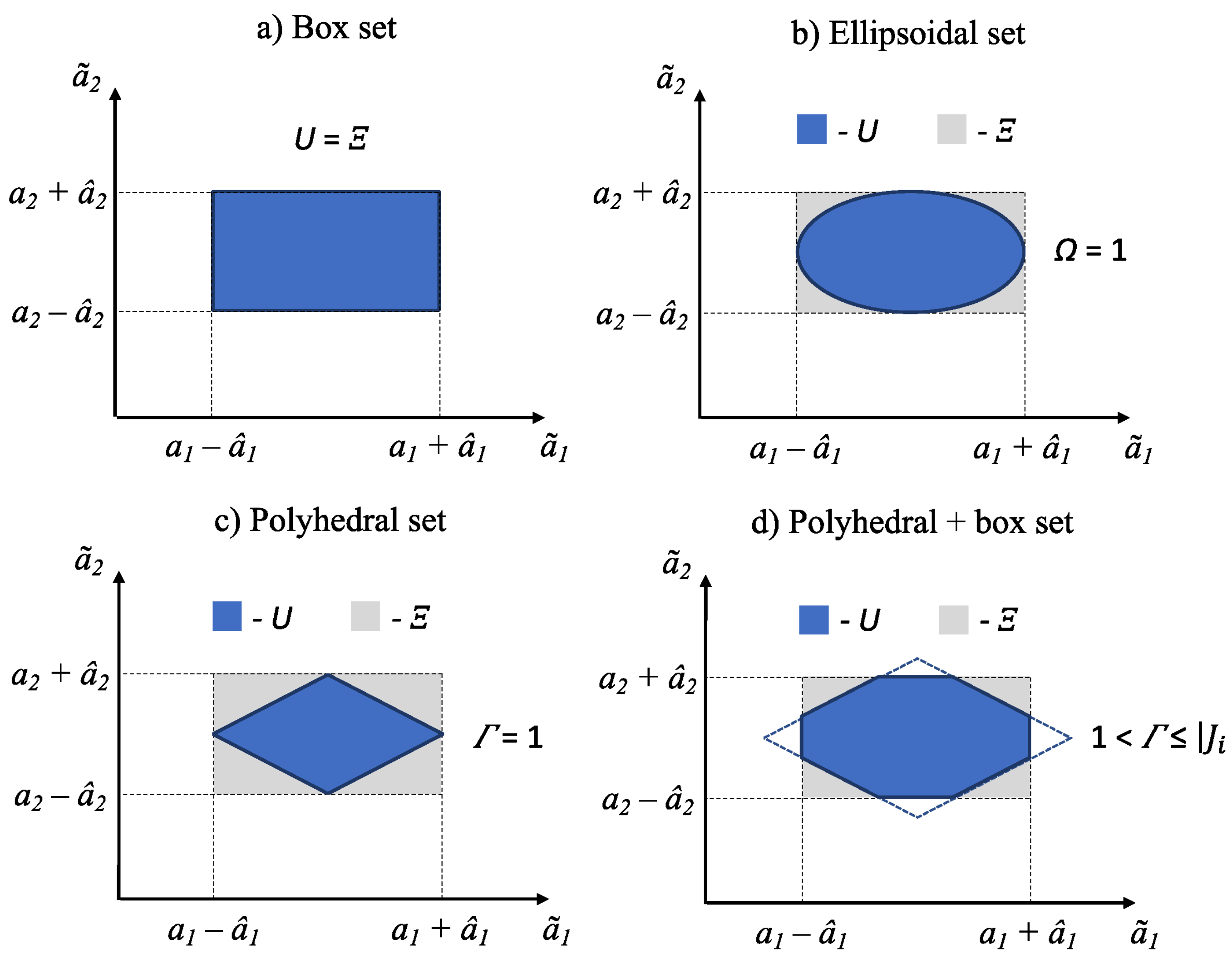
Design of uncertainty

With the available data past realisations or distribution hypothesis within the known mean, min and max values, an uncertainty set with radius/dimension Γ based on the desired risk level (`Prob` $\in [0,1]$) can be designed. Note that ϵ = 1 - `Prob` in the Figure below.



Then, the most common "geometric" form of the sets to be chosen with are (in order of conservativeness):

- Box
- Ellipsoidal
- Budgeted/polyhedral



Future development

For this tool it is not of actual interest to impose tight limitations to the computational time, since there is not the need to perform this optimisation *online* during in-plant operations. For this reason, it will be interesting to test a 'scenario approach' formulation of the formalisation of the robust optimisation problem (more intuitive, no need for random variables as decision variables, no need for uncertainty set dimensions' design, higher computational burden and time required).